Request For Proposal

Engagement of Agency for Automation of Operational Workflows and Centralized Data Management System for the National Accounts Division

Tender Reference No. L-11011/1/2024-NAD-10

Date of publishing: 01/09/2025

Government of India

National Accounts Division

Khurshid Lal Bhawan

Ministry of Statistics and Programme Implementation

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This Request for Proposal ("RFP") is issued by the Ministry of Statistics & Programme Implementation (MoSPI). The objective of this document is to solicit technical and financial offers from the parties interested in carrying out the scope of work as mentioned in this document. While this document has been prepared in good faith, no representation or warranty, express or implied, is or will be made, and no responsibility or liability will be accepted by MoSPI or any of their employees, advisors, or agents as to or in relation to the accuracy or completeness of this document and any liability thereof is hereby expressly disclaimed.

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Activities listed in this RFP to be carried out by MoSPI after the receipt of the responses are indicative only. MoSPI has the right to continue with these activities, modify the sequence of activities, add or remove activities, as dictated by the best interests of MoSPI.

Invitation

MoSPI invites Proposals/ Bids for onboarding an Agency for **Automation of Operational Workflows and Centralized Data Management System for the National Accounts Division** as per the scope of work defined in this RFP. The Bidders/Applicants desirous of taking up the project are invited to submit their Proposal/Bid in response to this RFP. The Bidders/Applicants should have the necessary experience, capability, and expertise to perform, as per the terms and conditions outlined in this RFP. The RFP is not an offer by MoSPI, but an invitation to receive responses from the potential Bidders. No contractual obligation whatsoever shall arise from the RFP process unless and until a formal contract or work order is offered by duly authorized official(s) of MoSPI with the successful bidder.

Factsheet

S.No.	Particulars	Details
1.	Assignment Title	Request For Proposal (RFP) for Engagement of Agency for **Automation of Operational Workflows and Centralized Data Management System** for the National Accounts Division
2.	Purchaser details	National Accounts Division (NAD), Ministry of Statistics and Programme Implementation (MoSPI), Government of India (Contact Details: 011- 23455429, Email: ddg3.nad@mospi.gov.in)
3.	Location	New Delhi, India
4.	Bid Submissions	Through GeM portal
5.	Method of Selection	Quality Cum Cost Based Selection (QCBS) **(Technical: 70, Financial: 30)**
6.	Consortium / Subcontracting	Not Allowed
7.	EMD	Rs. 60,00,000/- (INR Sixty Lakh Rupees only) in the form of Bank Guarantee.
8.	Performance Security/Bank Guarantee	5% of the contract/WO value discovered through tender process for this RFP
9.	Tender Availability	GeM portal and at the tender section of MoSPI website
10.	Date of publication of the RFP document	01.09.2025
11.	Pre-Bid Meeting	10.09.2025 (15:00 Hrs IST) at K.L Bawan, Janpath, New Delhi -110001
12.	Bid validity	180 days from last date of submission of the bid.
13.	Bid submission	As per GeM bid requirement
14.	Last date of submission of queries	08.09.2025 (15:00 Hrs IST) (As per attached format in **Annexure -2)** on GeM Portal only
15.	Language of bid	The bid should be submitted in English only

S.No.	Particulars	Details
16.	Last date of submission of bid	25.09.2025 (15:00 Hrs IST)
17.	Opening of Technical Bids	25.09.2025 (17:00 Hrs IST)
18.	Technical Presentation, Opening of Financial Bids & Issue of Work Order	To be intimated to qualified bidders.
19.	Duration of Contract	5 Years (2 years of development + 3 years of O&M)

Important Dates Table

S.No	Detail of Activities	Target Date and Time
1.	Date of publication of the RFP document	01.09.2025
2.	Last date of submission of queries	08.09.2025 (15:00 Hrs IST)
3.	Pre-Bid Meeting	10.09.2025 (15:00 Hrs IST)
4.	Last date of submission of bid	25.09.2025 (15:00 Hrs IST)
5.	Opening of Technical Bids	25.09.2025 (17:00 Hrs IST)
6.	Technical Presentation, Opening of Financial Bids & Issue of Work Order	To be intimated to qualified bidders

Key Definitions

1. Client means the Ministry of Statistics & Programme Implementation (MoSPI) who has invited bids for the project, with whom the selected Bidder signs the Contract for the Services and to whom the selected Bidder shall provide services as per the Terms and Conditions and Scope of Work of the contract.
2. Bidder means any entity registered under the relevant statute who have been invited to submit their proposals that may provide the Services to the Client under the Contract.
3. Day means calendar day.
4. Government means the Government of India.
5. LoI means the Letter of Intent issued by the Client to the selected bidder/Bidder.
6. Personnel means professionals and support staff provided by the Bidder and assigned to perform the Services or any part thereof.
7. Proposal means the Technical Proposal and the Financial Proposal.
8. RFP means the Request for Proposal prepared by the Client for the selection of the Bidder.
9. Assignment/job means the work to be performed by the Bidder pursuant to the Contract.
10. Onsite means the office of National Accounts Division, MoSPI, New Delhi

Acronyms

ADB	Asian Development Bank
AES	Advanced Encryption Standard
AMC	Annual Maintenance Cost
ASI	Annual Survey of Industries
PBG	Performance Bank Guarantee
CEA	Central Electricity Authority
CMS	Content Management System
COFOG	Classification of the Functions of Government
CSP	Content Security Policy
DES	Directorates of Economics & Statistics
DQ	Data Quality
DR	Disaster Recovery
EMD	Earnest Money Deposit
ETL	Extract, Transform, Load
FRS	Functional Requirement Specification
FSI	Forest Survey of India
GDP	Gross Domestic Product
GeM	Government E Marketplace
GFCF	Gross Fixed Capital Formation
GSDP	Gross State Domestic Product
GVA	Gross Value Added
GST	Goods and Service Tax
HCES	Household Consumption Expenditure Survey
HDFS	Hadoop Distributed File System

IMF	International Monetary Fund
IIP	Index of Industrial Production
IOTT	Input Output Transaction Tables
KVIC	Khadi and Village Industries Commission
LDAP	Lightweight Directory Access Protocol
MCA	Ministry of Corporate Affairs
MoA	Ministry of Agriculture
MoSPI	Ministry of Statistics and Programme Implementation
NAD	National Accounts Division
NAS	National Accounts Statistics
NIC	National Industrial Classification
NSDP	Net State Domestic Product
NSSO	National Sample Survey Office
ODATA/BAPI	Open Data protocol, Business Application programming Interfaces
PLFS	Periodic Labour Force Survey
PLI	Postal Life Insurance
PMO	Prime Minister’s Office
PoSB	Post Office Savings Bank
PSU	Public Sector Undertaking
RBI	Reserve Bank of India
RTI	Right to Information
RDBMS	Relational Database Management System
RBAC	Role based Access Controls
SAML	Security Assertion Markup language
SNA	System of National Accounts
SRS	Software/System Requirement Specification
SSH	Secure Shell
SSL	Secure Sockets Layer
SUT	Supply Use Table
TLS	Transport Layer Security
TRAI	Telecom Regulatory Authority of India
UAT	User Acceptance Testing
UNSD	United Nations Statistics Division
UT’s	Union Territories
VAPT	Vulnerability Assessment and Penetration Testing
WO	Work Order
WPI	Wholesale Price Index

1. Organization Background

The National Accounts Division (NAD) functions under the Ministry of Statistics and Programme Implementation (MoSPI), Government of India. The division coordinates with more than 300 data sources which includes central ministries, state departments, regulatory bodies, autonomous bodies, PSU's and various other stakeholders.

1.1. Roles and Responsibilities of NAD

a) Preparation of national accounts, which includes the estimates of Gross Domestic Product (GDP), National Income, Government/Private Final Consumption Expenditure, Capital Formation and Savings along with details of transactions of institutional sectors.
b) Bringing out annual publication 'National Accounts Statistics'.
c) Providing technical guidance and support to the State/UT Directorates of Economics & Statistics (DESs) on compilation and release of State Accounts, including estimates of State Domestic Product. State level estimates of GVA and GFCF in respect of supra-regional sectors, viz., Railways, Communication, Banking & Insurance and Central Government Administration.
d) Liaison with international organizations like UNSD, IMF, ADB, World Bank, etc. on national accounts matters.
e) Preparation and release of SUT and IOTT from time to time.
f) Setting up of Advisory Committee on National Accounts comprising eminent economists, statisticians and other experts, including Departmental representatives, to look into and advise on all methodological aspects of compilation and dissemination of national accounts, base year revision and back series calculations.
g) Discussions with the staff of IMF on GDP compilation as per guidelines under Article IV of the IMF Articles of Agreement.
h) Coordination with Ministry of Finance and other line Ministries/Departments, RBI, NITI Aayog, TRAI, Finance Commission of States and UTs, etc. on compilation and dissemination of National accounts statistics.
i) Handling RTI, Parliament Questions (Rajya Sabha & Lok Sabha), Parliamentary Assurances, VIP references, Cabinet Secretariat / PMO Matters, Complaints, Public Grievances, Budget matters including Outcome/Performance Budget, Audit and Court Cases related matters relating to the Unit/Division.

1.2. List of units at NAD

The following functional units are performing the afore-mentioned roles and responsibilities:

S.No.	NAD Unit	Functions
1	Agriculture	Estimates GVA and GVO for crops, livestock, forestry & fisheries. Gathers data from MoA, DESAg, FSI etc. Majority of data is received in Excel format with some being in PDF format.
2	Consumption of Fixed Capital, Gross Capital Formation, and miscellaneous estimations	The investments support production capacity and infrastructure, and are compiled using corporate filings, government budgets, and industry surveys.

3	Construction	Estimates GVA from construction activity. Utilizes data from private firms (MCA) and PSUs, additionally Integrates cement, steel, and investment indicators. Supports both annual and quarterly estimates.
4	Electricity, Gas, Water Supply and Remediation	Collects PSU data for electricity, gas, and water supply sectors. Sources include Central and State Departments, NSS Surveys, MCA data, the CEA, and autonomous bodies such as the KVIC. GVA and GVO are calculated at current and constant prices.
5	Gross Fixed Capital Formation	The GFCF Unit focuses on estimating capital investments made in fixed assets such as buildings, machinery, and equipment.
6	Manufacturing	Uses MCA and PSU data to estimate manufacturing GVA. Applies production approach and IIP for quarterly estimates. The unit utilizes external data sources such as the IIP and the WPI extracted through online portals. Shares with multiple units at NAD.
7	Mining and quarrying	Uses Indian Bureau of Mines for production and prices. IIP and mineral output support quarterly updates. Ensures sector accuracy. Subsequently, input rates are used to derive the GVA from the GVO. The data is received in both PDF and Excel formats.
8	AE/QE	Consolidates sectoral data for GDP estimation. Ensures SNA compliance and benchmarks with key indicators. Publishes Q1–Q4 GDP, aligns quarterly with annual targets, and maintains consistency for trend analysis. The data is organized on a quarterly basis by breaking down the annual estimates from previous years to project the values for the current financial year.
9	Non-Financial Non-Departmental Enterprises	Extracts financial data from around 2,200 Public Sector Undertakings (PSUs) for estimating GVA, investment, and savings. Sources data from Annual Reports and the Department of Public Enterprises survey. Uses standardized Excel-based toolkits to process and validate the data. Collaborates with the Non-Financial Private Corporation Unit to support national accounts compilation and prepares state-wise PSU estimates for integration into the economic accounts.
10	Financial Corporation	Collects financial data from RBI, NABARD, MCA, banks, insurance firms, and cooperatives. Uses standardized toolkits for analysis. A total of 292 variables are accounted, based on SNA codes. Currently working on excel based toolkits (VBA). CFC data is taken from PoSB and PLI from Savings unit. Provides data to the following units: SDP, Savings, PFCE, DE, AE/QE.
11	General Government; Public Administration and Defense	Collects public sector finance data for GVA. Uses budget documents, public accounts. General Government covers the following components: Government Controlled NPISH, Centre and State Governments, Rural /ULB's,

		Autonomous Bodies, Social Security Funds, Cantonment Boards. Economic codes. Purpose codes are mapped according to the classified items. MS Access and Excel are used to process the data in the unit.
12	Non-Financial Private Corporates (NFPC)	Processes MCA data (MGT-7/7A forms). Uses SPSS and manual validation. Supports GVA estimation in multiple sectors.
13	State Domestic Product (SDP)	Prepares GSDP for supra-regional sectors (Railways, Central Govt, etc.). Allocates national data to states. Uses IRASS, zone-level reports. Applies Labour Input method for informal sector.
14	Private Final Consumption Expenditure Unit	Estimates household and NPISH consumption. Coordinates internally for data on output, imports, and use. Computes PFCE at constant/current prices using Excel.
15	Non -Financial Services	Compiles GVA and GVO for various service sectors including real estate, legal, IT, and professional services. Estimates imputed rent for owner-occupied dwellings using market rental rates and housing stock data sourced from NSSO, Census of India, and municipal records. Also handles sectors like Trade, Repair, Hotels & Restaurants, Transport, Communication, Storage, and Broadcasting by collecting data in Excel/PDF formats from relevant ministries.
16	National Account Statistics	The NAS Division prepares key economic data like GDP, savings, and investments. It collects and compiles data from different sectors (government, households, companies) and ensures all estimates follow global standards. This helps in understanding the country's overall economic performance.
17	Savings	The Savings Unit is responsible for estimating savings across different institutional sectors, including households, private corporations, and the general government. It gathers data from various sources such as RBI reports, budget documents, MCA filings, and NSSO surveys. The unit calculates both gross and net savings in line with the SNA framework and supports macroeconomic planning by analyzing sector-wise savings trends.
18	Supply and Use Tables	Compiles Supply-Use Table annually. Uses toolkit with validations for consistency. The Supply and Use data must be aligned and any imbalances are adjusted to ensure consistency. Majority of data is received in Excel format and GST data is being downloaded in PDF format through online channels.

1.3. Existing Ecosystem and Processes

The data at NAD is sourced from annual/quarterly datasets, periodic benchmark surveys, central bodies, DES and other sources. Data is received from multiple sources (in different formats e.g. PDF/Excel/Scanned Documents/CSV etc.) having variety of indicators and parameters. The data is sanitized and cleansed by converting it into standardized format. The estimation and compilation are done afterwards based on the three approaches (production, expenditure and income). The key outputs are generated in the form of National Account Statistics (NAS), annual and quarterly estimates, state level GSDP/NSDP/Per-capita income, Supply use Tables and Input-Output Transaction Tables. Data integration and interoperability between multiple units at NAD help in seamless analysis and compilation of GDP estimates. The data is disseminated via press releases, data portal and various publications released by MoSPI.

It is to be noted that, in course of time MoSPI is planning to revise its base year (2011-12) and tentatively 2022-23 will be taken as base year in the base year revision exercise. Therefore, sectoral estimates may have to be revised wherever possible according to the approach adopted.

The flowchart below is an illustration of the indicative data workflow for the National Accounts Division (NAD).

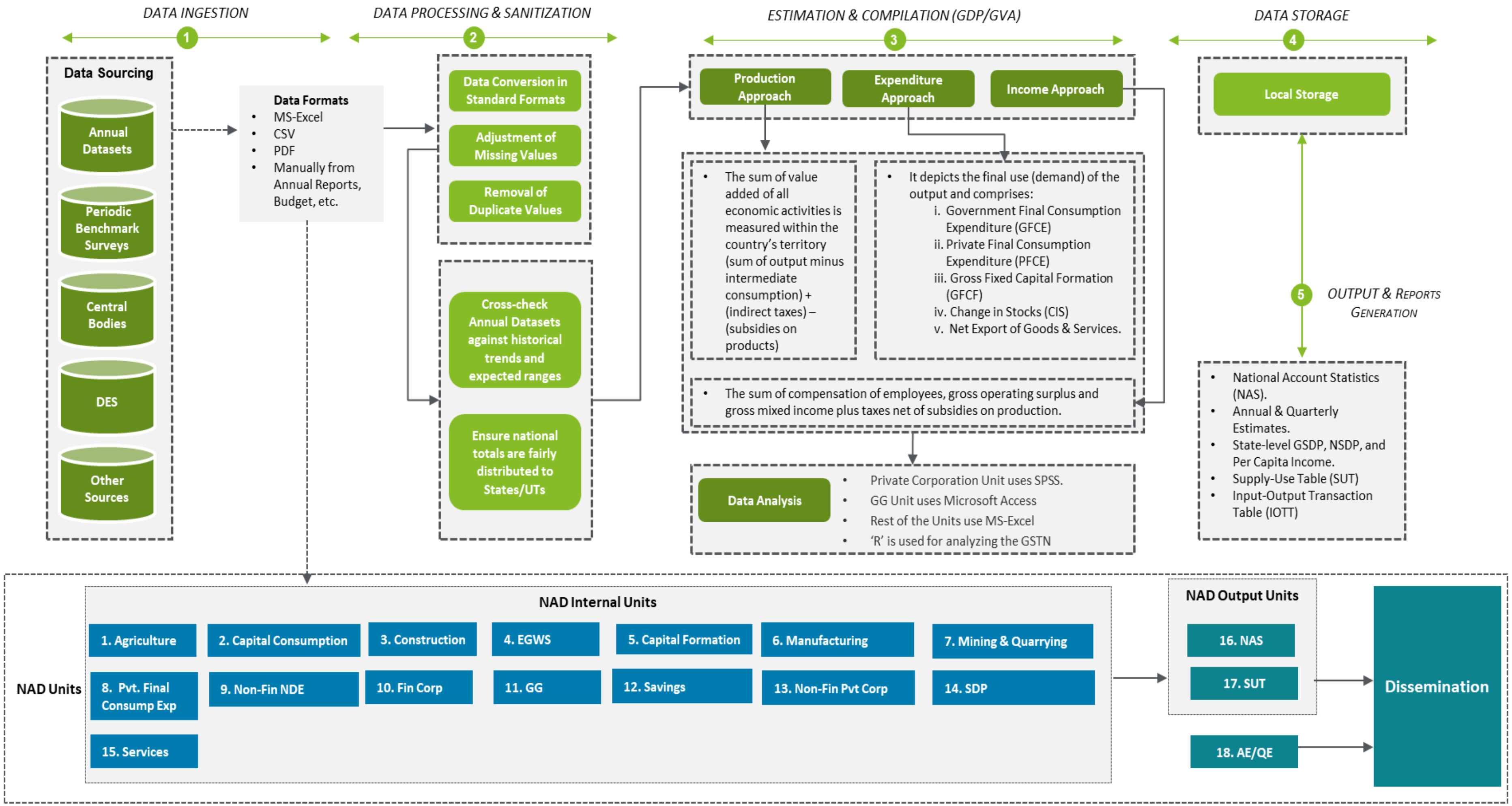

Figure 1: Indicative Data Workflow of NAD (1)

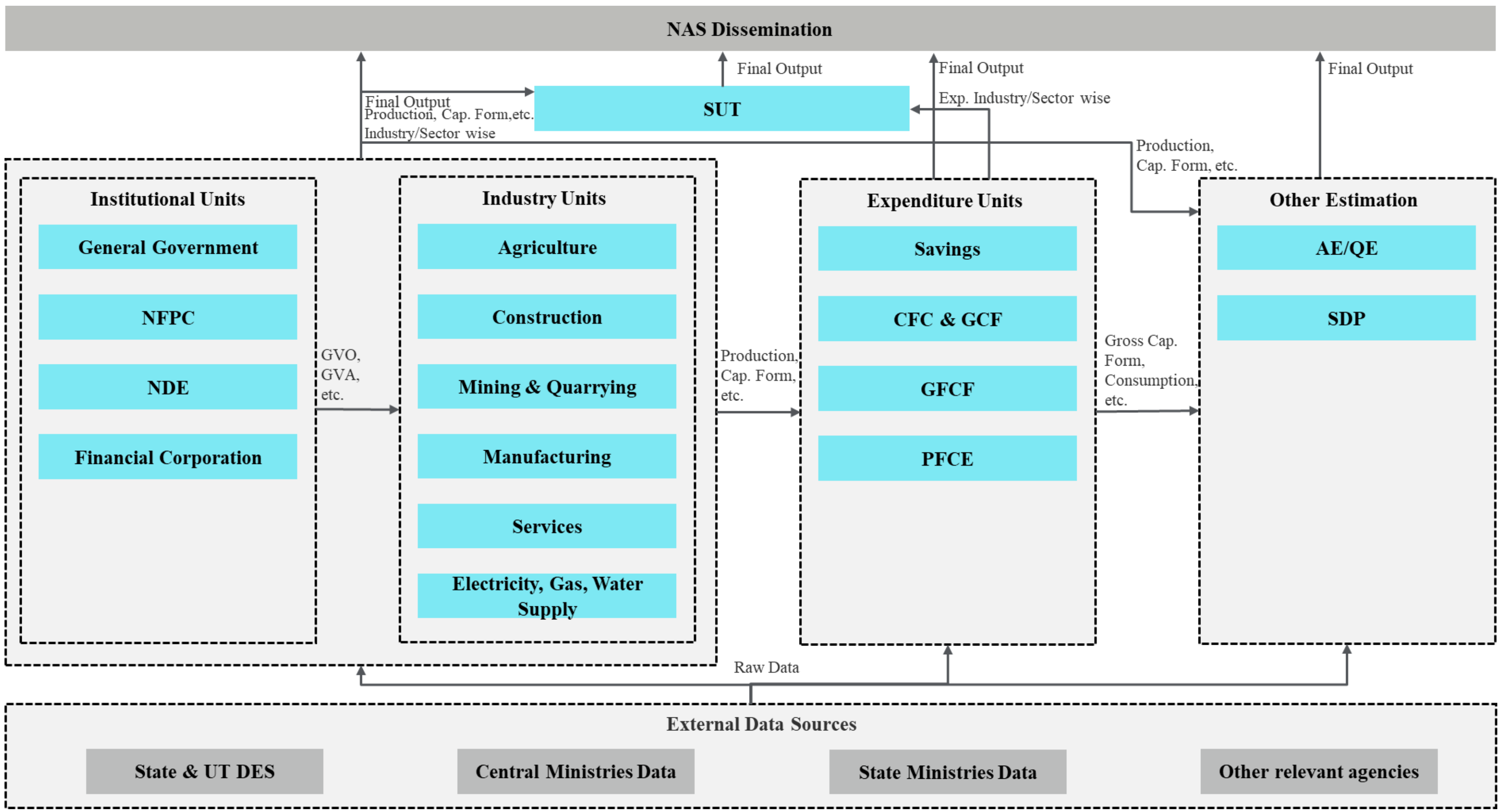

Figure 2: Indicative Data Workflow for NAD (2)

2. Objectives

a) Automated end to end workflows and processes – Design, Develop and Implement automation for the full data lifecycle—collection, ingestion, processing, validation, computation (e.g., GDP aggregates), and dissemination—removing the need for manual interventions.
b) Implement modular and scalable architecture – Develop solution with combination of smaller, independent, and interchangeable modules that can be tested individually. The proposed solution shall enable the system to handle increasing load, traffic or data without compromising performance.
c) Establish a unified, centralized and secure data platform – Develop and Implement a centralized data repository hosted on the Government Cloud that consolidates historical and current datasets from all sources **(Attached in Appendix-1)**.
d) Enable secure data sharing and RBAC – Implement Identity Access Management (IAM) and resource server that enables secure data sharing for data entry, validation, compilation, and analytics, configurable at dataset, department, dissemination levels etc. for different user roles.
e) Support interoperability across Ministry and Systems – The system should be able to interact and ingest data from internal and external stakeholders and designed accordingly.
f) Integrated analytics and monitoring capabilities – BI tools and analytical dashboards should provide customizable (automated report generation in various formats) reports and track key indicators both at micro and macro level.

3. Scope of Work

The engagement focuses on developing an integrated automated system for seamless compilation, processing and validation of data for GDP calculation while maintaining the highest standard of accuracy and compliances. The selected Bidder will be responsible to gather user requirements, design, develop, implement, operate, and maintain the automated system, its modules and dashboards while maintaining data inter-linkages with multiple stakeholders. There will be two phases of the project: **The development Phase will be a total of 24 months followed by 36 months of O&M.**

3.1. Functional Requirements

Based on the assessment carried out by the Bidder, FRS and SRS would be prepared regarding each unit at NAD supported by wireframes and mock-ups (4-5). The document would require comprehensive understanding of requirements at Micro and Macro level to attain the desired output.

3.1.1. Data Ingestion, Collection and Storage

The following points (but not limited to) are required to be addressed by the selected Bidder under this engagement:

a) Supports importing data from Excel, CSV, PDF, APIs, and web scraping.
b) Enables automated, secure, element-based data collection from internal and external sources, including XBRL, emails, file uploads, and web portals.
c) Performs regular web crawling and uses configurable parsers for structured and unstructured data.
d) Applies hygiene checks and maintains audit trails for traceability.

e) Maps all incoming data to a centralized database designed for efficient querying, analysis, and reporting.
f) Utilizes modular, low-code/no-code components for onboarding new data sources.
g) Supports varied protocols and allows flexible augmentation or removal of data sources.
h) Employs optimized storage mechanisms with compression for different use cases.
i) Includes a centralized cloud repository (data lake, warehouse, and data marts) with bidirectional data pipelines.
j) Designs scalable storage layers and ETL pipelines for raw and processed data.
k) Migrates legacy data (Excel, SPSS, Access) in bulk with automated validation and cleansing.
l) Handles diverse data formats from standard databases and maintains a data inward register.
m) Ensures high performance and reliability of the database infrastructure.

3.1.2. Data Processing

The following points (but not limited to) are required to be addressed by the selected Bidder under this engagement:

a) Accurately computes macroeconomic aggregates using NAD's standard methodologies, which involve complex, cross-functional relationships.
b) Supports data handling across multiple frequencies (monthly, quarterly, bi-annually, annually) and variables (indicators, categories, classes).
c) Applies configurable logic including lookup tables, formulas, algorithms, and mathematical calculations tailored to NAD unit requirements.
d) Automates dataflows from source systems to the data warehouse, with capabilities to detect schema changes and assess downstream impacts.
e) Enables real-time anomaly detection and alerting via operational dashboards.
f) Integrates advanced big data technologies such as Text Mining, Natural Language Processing (NLP), and Artificial Intelligence (AI).
g) Supports high-performance text analytics for processing large volumes of data efficiently.
h) Deploys a staging layer within a big data environment for scalable ingestion and processing.
i) Designs and delivers a robust Data Lake architecture alongside the conventional data warehouse, leveraging expertise in big data implementations.
j) Supports open-source tools like R, Python, and other interoperable applications for Data Lake operations.
k) Implements data masking for confidential elements as identified by NAD, ensuring privacy and security.
l) Provides a user-friendly, screen-based interface for metadata creation, editing, and management.
m) Includes a configurable data input module allowing NAD officials to update processing logic as needed.

3.1.2.1. Types of file processing required at NAD:

a. Excel/CSV File Processing

i) Support .xls, .xlsx, and .csv formats.
ii) Automatically detect and ingest data across worksheets and multiple files.
iii) Provide preview and edit functionality before final ingestion.

iv) Identify items, categories, and parameters using predefined formulas and Excel toolkits.
v) Enable batch processing for large and multiple files.
vi) Convert existing Excel workflows (macros, formulas) into ETL automation scripts.
vii) Integrate business process re-engineering into the Data Warehouse using modern architecture.
viii) Preserve NAD’s toolkits, formulas, and macros in an automated manner.

b. PDF File Processing

i) Text extraction capabilities of the system must preserve the logical flow of information while handling varied PDF layouts including, multi-column formats, headers, footers etc.
ii) The tables should re-construct exactly along with text positions as the source document.
iii) The extraction should be able to identify commas, separators, spaces from the source text to maintain maximum accuracy and data integrity.
iv) Quality assessment mechanism should alert the user on low-confidence extractions for manual review.
v) Use OCR and AI tools to digitize incoming PDFs, auto-parse field values, and flag inconsistencies.
vi) A preview and edit feature should be provided before the final insertion is made.

c. Email

i) The proposed automation solution shall include a robust RPA component specifically designed to ingest, parse, and extract structured and unstructured data from inbound emails and associated attachments.
ii) The RPA robots must interface seamlessly with the NAD’s email systems, apply configurable extraction rules, perform validation, and feed cleaned data into downstream applications or databases.

d. Web Forms

i) Form data extraction involves identifying and extracting information from structured forms including checkboxes, radio buttons, text fields, and signature blocks. The system should handle both filled and unfilled forms and maintain the relationship between form field labels and their corresponding values.
ii) The integration between the system and input from web forms should correspond to standardised format as per requirement of each unit in NAD.

e. API Integrations

i) Allow preview of data before API-based insertion into the database.
ii) Securely store API credentials and maintain audit trails for all access.
iii) Implement rate limiting and error handling to ensure data integrity and prevent overload.
iv) Automate data ingestion from Excel and email attachments with retry mechanisms.
v) Use a metadata-centric ETL approach to capture source/target structures, support impact analysis, and manage schema changes.
vi) Reimplement Excel macros/formulas (e.g., price estimation) in backend code or database procedures.
vii) Schedule ingestion and processing tasks using cron jobs or Airflow DAGs.

viii) Log all steps (success, failure, data quality) and generate dashboards or reports.
ix) Use Git for version control and automate script testing and deployment.
x) Maintain an Operational Data Store (ODS) for real-time reporting with frequently updated snapshots.
xi) Support archival of structured/unstructured data, source files, data warehouse, ODS, data marts, reports, and dashboards.
xii) Support of providing a south bound interface through which any data producing agency can ingest data using their interface.

The system should be integrated with the following tasks (not limited to) that are required for processing of data by various units:

a) Interlinkages of Input and Output Calculations within units of NAD should be maintained into the new system.
Indicative Example: Unit 1: Manufacturing unit, Construction Unit, SDP Unit. Fixed capital data is compiled by manufacturing unit and shared with construction unit for their input and review. The data from construction unit is shared back to manufacturing unit with their input which in turn is incorporated by SDP unit as their output.
b) The system should be able to assign data on all India level data to individual states based on specific ratios used for calculation by certain units of NAD.
c) All toolkits (Excel, MS Access, SPSS) should be transported to the new system without discrepancies and error. All these toolkits have to be integrated together for input and output of data by various units of NAD.
d) Output for one unit can be the input for another (Back and forth movement).
e) The system should have a structured and well-defined system to integrate GST data which is analyzed using the (R) programming language and MCA data which is processed in SPSS software.
f) The system should be able to receive datasets from external agencies though a standardized format to interact with the units at NAD.
g) The portal shall include an integrated analytical toolkit module to facilitate the analysis of NSO survey datasets such as ASUSE, PLFS, AIDIS, HCES, etc., for use in GVA compilation.
h) The selected Bidder shall provide a comprehensive Toolkit for Survey Data Analysis as part of the project deliverables. This toolkit must enable efficient ingestion, processing, analysis, and visualization of survey data collected from multiple sources, including structured, semi-structured, and unstructured formats.
i) Provision of balancing interface is required in SUT datasets. The system should also integrate tools to edit unbalanced SUT's to achieve balanced tables. (Attached in Annexure -1)

3.1.3. Data Validation

The following points (but not limited to) are required to be addressed by the selected Bidder under this engagement:

a) The system should detect changes in source schemas, propose necessary modifications, and preserve backward compatibility during transitions.
b) The system should ensure detection and resolution mechanisms for duplicate field and error handling.
c) The portal should provide data export services (e.g. bulk downloads) and implement rule-based checks related to any anomalies detected.
d) Automated alerts should be sent if validations fail, or data thresholds are exceeded.

e) Incremental loading should use change detection mechanisms such as timestamps, version numbers, or hash comparisons to identify modified records.

f) Error recovery should include the ability to resume failed loads from checkpoints and handle partial failures gracefully.

g) The Bidder is required to document, implement and enhance all the validations currently being used in Excel and other statistical toolkits.

h) Analyse the existing Excel-based toolkits (which include built-in formulas, macros and data validation for GDP estimates, price adjustments, etc.) and implement equivalent computation modules.

i) The key outputs derived in NAD are GVA and GVO at current and constant prices. For example, MoSPI's GDP methodology uses price indices to compute constant-price estimates ("estimates at constant prices are arrived at by deflating the current price GVA estimates by suitable price indices"); these deflation formulas must be encoded into the new system so that all current/constant price calculations continue seamlessly.

j) Preserve all critical calculations and checks from the spreadsheets.

k) Implement workflow automation to eliminate manual handoffs. Replace email/Excel forms with standardized web forms or direct API submissions for data collection processes.

l) Automate notification and approval steps so that data, once submitted, triggers automatic validation routines and alerts of missing values or discrepancies.

m) The system should deal efficiently with duplicate data. Recommendations should be given for identifying the duplicate data and ways for terminating its further processing.

n) The architecture should cater to efficient addition and removal of tags for each individual data element. Each data element can have multiple tags for easy searchability. Apart from auto-tagging, there should be a facility to tag incoming ingested data elements manually. Existing data elements can also be searched and re-tagged manually.

o) Users should have the ability to generate and download an "Offline Data Cube" containing all data series they are authorized to access. This offline package should be installable via a menu option, allowing users to copy it to their laptops for use without internet connectivity.

p) The following validation points (but not limited to) would require in-depth analysis to maintain uniform data structure across datasets:

i) **Unit Metrics:** Data received from various state department and external sources may not use the same metrics as NAD. (E.g.: State department might share data in thousand lakh and NAD uses crores as the currency figure to depict prices). The system should be able to have a standardized template of automatic conversion when the source data is ingested into the system.

ii) **Filtration & Selection:** There might be multiple indicators used within the division and externally which won't be required by a particular unit to carry out a certain operation. The unit should have a parser to filter out the indicators/categories required to be used for that particular task.

iii) **Conditional Formatting**: As an example, NULL column should automatically populate data based on logic provided beforehand to avoid any discrepancy. (e.g. Average of last 3 years can be taken as an input for the "Null" cell value as an estimate). Also, a manual editing feature should be accompanied, to tweak input as per unit requirements.

3.1.4. Metadata Management

The selected Bidder will be accountable for the design, development, and deployment of a robust Metadata Repository as an integral component of the proposed solution. This repository will function (not limited to) as a centralized, well-structured platform for the storage, management, and retrieval of metadata associated with all datasets, processes, reports, and data sources within the project's operational framework.

a) The repository shall be developed using industry-standard metadata frameworks (e.g., ISO 11179, Dublin Core, or equivalent) and must support custom taxonomies, data dictionaries, and classification schemes as defined by the Client.
b) The system must support capture and maintenance of the following categories:
 i) Technical metadata (data types, data formats, schema, lineage, etc.)
 ii) Business metadata (definitions, ownership, usage policies)
 iii) Operational metadata (data processing history, refresh cycles, logs)
 iv) Security and access metadata (data classification, sensitivity, role-based access).
c) The metadata repository must be able to interact with existing data storage and processing systems to automate metadata extraction (e.g., from databases, ETL pipelines, APIs, or Excel files).
d) The solution must provide intuitive search and browse functionality for users to discover metadata assets and their relationships. Visualization of data lineage and relationships is preferred.
e) The repository should include mechanisms for metadata approval workflows, audit trails, version history, and role-based access to ensure compliance and governance.
f) The suggested solution shall incorporate a strong Business Glossary feature within the metadata management framework. The Business Glossary will act as a centralized resource for domain terminology and definitions utilized throughout the organization. It must support role-based ownership and stewardship, version control, and audit trails to ensure accountability and traceability. The glossary should be integrated with metadata repositories and data assets, allowing linkages between business terms and underlying technical components such as databases, reports, or APIs. It should also include features for search and discovery, support for synonyms and classifications, and enable collaborative workflows for review and approval.
g) The metadata repository must comply with relevant data governance and interoperability standards mandated by the Client or applicable government regulations.
h) The Bidder shall provide complete technical and user documentation, along with training for relevant stakeholders on how to maintain and utilize the metadata repository.

3.1.5. AI/ML Integration

a) The solution must include AI/ML capabilities to automatically identify, extract, and structure data from diverse sources (e.g., scanned documents, PDFs, spreadsheets, APIs, and unstructured text).
b) AI-based algorithms must be used to classify incoming data by topic, source, or domain, and apply relevant tags or metadata for efficient organization and retrieval.
c) AI models shall be deployed to assess data quality, detect anomalies, missing values, and inconsistencies, and flag outliers or suspicious patterns for review.
d) The system must support both rule-based validation and machine learning models that learn from historical data to predict and flag invalid or unlikely values.
e) The platform should use AI to automate transformation rules (e.g., unit conversion, format standardization) and intelligent mapping between disparate schemas or formats.

f) AI techniques should be employed to reconcile data across multiple sources, detecting duplicates and merging records with similar attributes.
g) The AI components should have the ability to learn from feedback and historical data, improving the accuracy and efficiency of ingestion and validation over time.
h) All AI-driven actions must be transparent, with detailed logs explaining how a decision (e.g., rejection, classification, flagging) was made.
i) AI modules must seamlessly integrate with the organization's existing data lakes, warehouses, ETL tools, and dashboards through APIs or connectors.
j) The proposed solution must provide Natural Language Processing (NLP) and Generative AI capabilities and ensure AI readiness for use cases identified during the development and O&M phase of the project. Few indicative use cases are outlined below:
   i) Data Validation, Data Ingestion, Forecasting of the indicators/Nowcasting.
   ii) Data extraction from documents.
k) The system should incorporate a feedback mechanism allowing users to validate or correct AI outputs, which in turn should be used to refine models.

3.1.6. Conversational AI Chatbot

The selected bidder shall develop and execute a secure, web-based conversational AI chatbot solution that should be available to authorized internal users (NAD, officials, statistical officers, data entry operators) and must comply with the following requirements and deliverables:

a) The selected bidder will create and train the chatbot's NLP model utilizing the data and information through historical datasets and information presently available at NAD. The chatbot should comprehend concepts related to GDP calculation and economic indicators, and be adjusted with business-related data (e.g., GDP compilation rules, definitions of indicators) to provide accurate, context-sensitive answers. The outcomes should consist of a trained dialogue model and relevant training data sources.
b) The selected bidder will connect the chatbot to the organization's data ingestion system. It should accept inputs from Excel files, databases, outputs from statistical tools, scanned PDF files, and OCR-processed physical documents. The chatbot will link to both structured and unstructured sources to retrieve pertinent data and initiate automated workflows when necessary. The solution should enable effortless data retrieval and computations fundamental to the GDP process.
c) All chatbot interactions and data access must be restricted by user role. The system shall record an audit trail of queries and actions for accountability.
d) The selected bidder shall comply with all organizational IT security standards, including encryption in transit and at rest.
e) The system shall record an audit trail of queries and actions for accountability. The selected bidder shall comply with all organizational IT security standards, including encryption in transit and at rest.
f) The chatbot shall provide multilingual NLP support (language identification and translation) when necessary to cater to various languages.
g) The selected bidder will be accountable for timely delivery of the complete chatbot system along with post-implementation support and maintenance as outlined in the contract.
h) The system should be created in such a way that it can be retrained on the latest data.
i) The chatbot shall implement appropriate guardrails and queries only related to official statistics shall be answered.

3.2. Functionalities of Dashboard and Web-based Portal

The data ingestion dashboard/portal would serve as the primary command centre offering (not limited to) processing of statistics, generation of reports & visualizations and display indicators for data validation & flagging of alerts. The dashboard should provide pages for intuitive viewing, querying and visualizations based on the content inserted by NAD officials while working on variety of datasets. The proposed solution should continuously evolve to meet new and changing user information needs, keeping in mind visualization, display, alert and user self-generated reports wherever considered feasible.

3.2.1. Collaboration and Version Control

a) The platform shall support simultaneous access by multiple users working on the same datasets while maintaining data consistency and preventing conflicts. The system must implement real-time synchronization mechanisms.
b) Every modification made to datasets within the portal must be tracked and stored with detailed metadata including user identification, timestamp, change description, and affected data elements.
c) The version control system should maintain complete historical records of all data transformations, calculations, and analytical operations.
d) Users must be able to compare different versions of datasets, identify specific changes between versions, and restore previous versions when necessary.
e) The system should provide clear visualization of change histories and support branching capabilities for experimental analyses that may later be merged with main datasets**.**

3.2.2. Built-in Statistical tools

a) The portal must incorporate comprehensive statistical analysis capabilities that replicate and enhance the functionality currently provided by SPSS, Excel, MS Access and other statistical software packages.
b) These tools should include descriptive statistics, inferential testing, regression analysis, time series analysis, and advanced econometric modelling capabilities.
c) The statistical computing engine should provide both guided workflows and advanced programming interfaces for statisticians.
d) The system must support standard statistical programming languages and provide extensive libraries of pre-built analytical functions commonly used in national accounts compilation.

3.2.3. Automation Controls

a) User should be able to manage the interface adapting to changing business requirements without requiring developer intervention**.**
b) The system should manage data schedules (e.g. data received monthly, quarterly, annually) and integrate into the portal seamlessly. This will allow manual data entry processes transform into automated workflows that run according to business requirements rather than staff availability.

3.2.4. User Management and Role based access

a) RBAC should align with NAD’s organizational structures and job responsibilities, providing each user with exactly the access they need without overwhelming them with unnecessary options or exposing them to inappropriate system functions.
b) The system should support hierarchical role structures, unit wise structures and file-based structures that reflect organizational reporting relationships, enabling NAD’s

officials to access information about their team's activities while restricting access to sensitive functions or data outside their purview.

c) Admin roles should be clearly defined along with provision to add/remove existing roles and add new ones.

3.2.5. Object Based Access Control (OBAC)

The suggested solution should incorporate Object-Based Access Control (OBAC) to apply detailed, flexible access control policies at the object level. The system must enable the definition and management of access permissions for particular data entities, reports, APIs, or application components according to user roles, attributes, and object metadata. The following points (not limited to) must be adhered in the proposed system:

a) Enable administrators to set object-level permissions through a unified interface.

b) Facilitate flexible policies reliant on user and object characteristics (e.g., division, information classification).

c) Record all access events for oversight and evaluation.

d) Ensure that unauthorized individuals are unable to access, alter, or execute prohibited objects.

e) Incorporate with Identity and Access Management (IAM) allowing detailed control of access to: Single datasets Modules or APIs for applications, Interface components (e.g., buttons, forms), Deployment assets (e.g., environment, build pipelines).

3.2.6. Database Access Management Tool (DAMT)

The Bidder will provide and implement a comprehensive Database Access Management Tool (DAMT) that consolidates control, oversight, and auditing of all database access throughout the NAD’s ecosystem. The DAMT should incorporate (not limited to) the following features:

a) Assistance for access policies based on roles and attributes along with detailed permission management at the schema, table, row, and column levels.

b) Storage of all database credentials, SSH keys, and certificates in an encrypted form

c) Record complete SQL queries for all activities conducted by privileged users.

d) Unchangeable, tamper-proof audit logs for all access requests and activities.

e) Prebuilt connectors for RDBMS system for integration and interoperability.

3.2.7. Content Management System (CMS)

The portal will be managed and controlled through the ‘Application Admin’ module and ‘Content Management’ module of the Portal with an intention of making most of the information available for stakeholder’s consumption through the web-based portal. The selected Bidder shall develop a comprehensive, browser-based CMS with information control and display features, enabling any NAD user to access and operate the system seamlessly from any location.

3.2.8. Multilingual Support

The portal should be well equipped to support Hindi and English language. The data ingestion & dissemination both require the system to read the language formats and map the data accordingly.

3.2.9. Advance Search Filters

The portal should support search and filter functionality based on predictive search, keyword-based search and dynamic viewing options for queries.

3.2.10. Security and Authentication

a) Support single sign-on (IAM) or a robust login system.
b) Access controls must restrict data visibility by role/unit.
c) All data in transit and at rest must be encrypted.
d) Logging should audit who accessed or changed data. The system should guard against common web threats (SQL ingestion, XSS) per ISO/IEC 27001 best practices.
e) Audit capabilities track all user activities within the system, providing detailed logs of who accessed what information when, and what actions they performed.
f) The Bidder shall monitor and provide the following security layers:
    i) Layer 4 - Layer 11 Firewall
    ii) Intrusion prevention / detection (network and host level)
    iii) Anti-virus, event logging and correlation
    iv) Vulnerability protection through implementation of proper patches and rules
    v) Vulnerability Assessment and Penetration testing before Go-live of any module
    vi) Database Activity Monitoring
    vii) Multi-Factor Authentication
    viii) SSL VPN
    ix) Anti-DDoS
    x) Data Leakage Prevention

3.2.11. Guiding Features and Design Principles

The following design features (not limited to) collectively would ensure a comprehensive, user-friendly, and efficient platform for managing innovation challenges and fostering a collaborative environment.

a) **Low-code / No-code:** Low-Code or No-Code are methods of designing and developing apps using intuitive drag and drop tools that reduce or eliminate the need for traditional developers to write code. The Selected Bidder should take this approach for the proposed work.
b) **Modular:** The portal should follow micro-service and event-driven architecture practices with each module of the application catering to specific business/ technical use cases.
c) **High availability**: The portal should be highly available and rapidly scalable based on the traffic. Minimum concurrency checks should be tested for activity-based requirements.
d) **Distributed**: The components of the portal should be distributed in a multi-server environment, enabling easier maintenance and reliability.
e) **Policy-driven management**: The application deployment should be governed by well-defined usage and data policies, enabling stronger control on the portal. Policy driven decisions are required to be accounted for scaling and management of infrastructure, archival, maintenance and usage of data storage mechanisms, logs and metrics, security & compliances, accessibility and data retention & archival policy.

3.2.12. Frontend Functionality

The system should have the following features (but not limited):

a) The portal/dashboard should be easily navigable and searchable.
b) The portal/dashboard must be accessible to all the units in the NAD division having a clean and systematic layout of options.
c) Reveal additional content or options only when users interact with specific elements, reducing clutter and cognitive load.

d) Provide easy-to-use buttons or widgets to share data insights and validations while working on worksheets on the portal/dashboard.
e) Deploy a clean, modern, and visually appealing design aesthetic that is consistent across all units of NAD, fostering a positive systematic and user-friendly experience.

3.2.13. Backend Functionality

The system should have the following features (but not limited):

a) NAD's database design, architecture, structuring, and coding focusing on database tables, views, back-end logic, application programming interface (APIs), performance, server APIs, repositories, frameworks ensuring the website performs correctly and along with being scalable.
b) The solution should have provision for an API Gateway to offer a single-entry point for defined back-end APIs and microservices (which can be both internal and external).
c) Utilize asynchronous processing techniques to handle long-running tasks without impacting user experience or application responsiveness.
d) Integrate real-time analytics tools to monitor user activity, platform performance, and identify areas for improvement.
e) Maintain detailed logs of user activity and system events for security purposes, troubleshooting, and compliance with regulations.
f) Implement a comprehensive monitoring system that generates alerts for potential issues or performance bottlenecks, allowing for proactive intervention and maintenance.
g) Notifications should be sent to the concerned stakeholders on scheduled events, reminders and flagging w.r.t sharing of particular dataset or discrepancy encountered in the data.
h) Establish a comprehensive data backup and recovery strategy to ensure quick restoration of data in case of system outages or data corruption.

3.2.14. Administrative Functions

a) Global parameter management allows administrators to set system-wide defaults that apply across all data processing activities, while still permitting specific overrides for unique requirements. These tests should simulate actual data retrieval operations to identify potential issues such as authentication failures, network connectivity problems, or changes in data source structure.
b) The web-based portal must function consistently across all major web browsers including Chrome, Firefox, Safari, and Edge.
c) The user interface should be responsive and adapt seamlessly to various screen sizes.
d) The system must provide consistent functionality across different operating systems and should not require users to install additional software or browser plugins.
e) The system should automatically archive older logs and provide retrieval capabilities when historical analysis is needed.

3.2.15. Training and Capacity Building

a) The selected Bidder would be responsible for capacity building of all the stakeholders (including Ministry/State Officials/Teams and other stakeholders).
b) Refresher training during release of each new version/upgrades and dashboard to NAD officials.
c) The training materials in the form of hard copy as well as soft copy need to be created, generated, maintained, and provided by the selected bidder.

d) From each unit, 5 to 10 officials shall be trained on how to use the module and shall be involved in testing phase of NAD's Data Ingestion Dashboard.
e) It will also be required to have historical dataset to ensure enough test cases are run to check integrity of the data ingestion dashboard.
f) Unit officials shall be testing the data ingestion dashboard before the actual go live-process.
g) In addition, tutorial and videos should be made for enhanced understanding of the officials.

3.3. Non-Functional Requirements

The system should incorporate a multi-tier security framework that includes (but not limited to) role-based access control, encrypted data transmission, secure audit trails, and limited access to preliminary estimates prior to their official release. It must adhere to the IT security guidelines prescribed by the Government of India, ensuring the use of approved encryption protocols and compliance with the Information Technology Act for managing sensitive government information.
The system should not compromise on any of the parameters listed below:

a) Security – The data should be stored in a secure environment.
b) Reliability – The data should not be lost in migration and processing certain outputs.
c) Accuracy – The data should be accurate and not reflect inconsistencies.
d) Integrity- The data should not be compromised anywhere throughout its lifecycle.
e) Flexibility – The infrastructure should be flexible enough to handle growing data requirements along with increasing data complexities.
f) Resilience – The developed framework should be resilient in nature, with high availability, disaster recovery, and backup/restore capabilities.

3.4. System Architecture and Implementation

a) The selected bidder will deploy the system on a secure cloud platform (provided by MoSPI). This includes provisioning servers (compute instances or serverless functions), database and object storage.
b) Data in transit and at rest must be encrypted – for example, use TLS for API calls and AES-256 for storage.
c) Structured data will reside in a relational or cloud data warehouse (supporting SQL queries), while raw data can be stored in a data lake/bucket.
d) The Bidder will develop back-end components in any language (REST APIs or batch processors) for data ingestion, validation, and retrieval. The environment will be containerized or virtualized (e.g. Docker, Kubernetes, virtual environments) to manage dependencies and facilitate updates.
e) Code will be managed in a repository with CI/CD pipelines to enable rapid deployment.
f) Provision cloud resources (database instances, storage buckets, compute) according to load estimates.
g) Configure VPC/network, secure firewall rules, and load balancers. Apply encryption settings (SSL certificates, encrypted volumes).
h) Create databases/data warehouses and configure object storage for raw files. Define backup and replication policies.
i) Implement programming-based language services for data processing, reading sources, performing transformations, writing to target stores. Include robust data profiling and validation as part of ETL. The ETL Program should have declarative design